

Decision-Making for a Non-Humanoid Tidying Robot: A POMDP Approach to Household Tidying Under Uncertainty

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Abstract—Household tidying presents a sequential decision-making problem under uncertainty, characterized by partial observability, stochastic actions, and dynamic environments. This work formulates tidying as a Partially Observable Markov Decision Process (POMDP) and implements lightweight belief-based policies for a simulated tidying robot. We compare Random, Greedy (perfect information), Basic POMDP, and Enhanced POMDP policies across grid-based environments with 80% detection accuracy and pickup success rates. Our Enhanced POMDP incorporates strategic belief initialization, exploration balancing, and spatial reasoning, achieving 11.9 points average reward, a 56.6% improvement over Basic POMDP and reaching 93% of the Greedy policy’s performance. Results demonstrate that computationally efficient belief-based methods can effectively handle uncertainty in domestic robotics applications while maintaining near-optimal performance.

I. INTRODUCTION

Household tidying, picking up toys, clothes, and miscellaneous items before vacuuming, represents a persistent daily challenge in domestic environments. This “pre-vacuum tidying” problem is universal yet remains largely unaddressed by current home automation solutions.

We formulate this as a sequential decision-making problem under uncertainty: a robot must navigate through environments, detect objects of varying priority, and decide on an optimal sequence of actions to maximize cleanliness while contending with imperfect perception and stochastic actions. The problem exhibits three key characteristics that motivate a POMDP formulation:

- **Partial Observability:** Limited sensor range and occlusion prevent complete environment perception.
- **Stochastic Transitions:** Grasping actions succeed with only 80% probability.
- **Dynamic Environment:** New objects may appear independently of robot actions.

This work applies POMDP planning to the domestic tidying domain, implementing and evaluating four policies. Our primary contribution is an Enhanced POMDP policy that achieves 11.9 points average reward: a 56.6% improvement over basic POMDP methods while maintaining computational efficiency suitable for real-time operation. The Enhanced POMDP achieves 93% of the performance of a perfect-information

baseline, demonstrating that near-optimal tidying is feasible under realistic sensing and actuation uncertainty.

The remainder of this paper is organized as follows: Section II reviews related work. Section III formally defines the problem. Section IV details our methodology. Section V presents experimental results. Section VI discusses implications and limitations, and Section VII concludes the paper.

II. RELATED WORK

This work builds upon three key areas of research: probabilistic robotics, POMDP planning, and domestic service robotics.

A. Probabilistic Robotics and POMDPs

The foundational textbook by Thrun et al. [1] established probabilistic approaches for handling uncertainty in robotics. For decision-making under uncertainty, Kaelbling et al. [2] formalized the POMDP framework, showing how belief states can represent uncertainty over world states. More recently, Silver and Veness [3] developed Monte Carlo Tree Search methods for large POMDPs, though their computational requirements often limit real-time applications.

B. Domestic Service and Cleaning Robots

In domestic robotics, Martinez et al. [5] implemented planning for surface cleaning tasks, though with simplified uncertainty models. Most recently, Wu et al. [7] demonstrated large language models for personalized tidying assistance, representing a different approach focused on high-level task understanding rather than low-level uncertainty handling.

C. Gap and Contribution

Existing work either: (1) develops theoretical POMDP methods with high computational cost, or (2) implements domestic robots with simplified uncertainty models. This work bridges this gap by applying lightweight, computationally efficient POMDP methods specifically to the tidying problem with explicit uncertainty parameters (80% detection/action success rates). We provide a comparative analysis of policy classes and demonstrate that heuristic enhancements to basic POMDP methods yield 73% performance improvement while maintaining real-time feasibility.

III. PROBLEM FORMULATION

We model the household tidying task as a Partially Observable Markov Decision Process (POMDP) defined by the tuple $(S, A, O, T, Z, R, \gamma)$. The robot operates in a discrete, grid-based home environment containing objects of different types—clothes, toys, and trash—that must be detected, retrieved, and removed. Uncertainty arises from limited perception and stochastic action outcomes.

A. State Space S

A state $s \in S$ encodes:

- the robot's grid position (x_r, y_r) ,
- the presence or absence of an object in each cell,
- the object type (if present).

Formally,

$$s = (x_r, y_r, \{o_i\}),$$

where $o_i \in \{\text{none, clothes, toy, trash}\}$. Because the exact environment configuration is hidden, the agent maintains a belief state $b(s)$ over possible world states.

B. Action Space A

The robot may execute:

- movement actions: up, down, left, right,
- pickup: attempt to grasp an object.

The simulator does not include an explicit drop action; objects are removed when successfully picked up.

C. Transition Model $T(s'|s, a)$

Transitions are stochastic due to imperfect actuation:

- movement actions are deterministic,
- pickup succeeds with probability 0.8.

A failed pickup leaves the object in the same location. Noisy clutter generation is modeled by objects randomly appearing in the initial state.

D. Observation Space O

At each step, the robot receives an observation of its current grid cell:

$$o \in \{\text{object detected, nothing}\}.$$

E. Observation Model $Z(o|s, a)$

Observations are noisy:

$$P(o = \text{object detected} \mid s \text{ contains object}) = 0.8,$$

$$P(o = \text{nothing} \mid s \text{ contains object}) = 0.2.$$

F. Reward Function $R(s, a)$

Rewards reflect the priority of different objects:

$$R(s, a) = \begin{cases} +5, & \text{successful pickup of trash,} \\ +3, & \text{successful pickup of a toy,} \\ +2, & \text{successful pickup of clothes,} \\ 0, & \text{movement or failed pickup.} \end{cases}$$

A final cleanliness score is computed after each episode.

G. Objective

The goal is to compute a policy $\pi(a|b)$ that maximizes the expected cumulative reward:

$$\pi^* = \arg \max_{\pi} E \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right],$$

with discount factor $\gamma = 0.95$. The robot must balance exploration (inferring hidden objects) and exploitation (cleaning visible clutter) under uncertainty.

H. Episode Termination

Episodes terminate when either:

- the robot's battery charge reaches zero, or
- a maximum of 50 time steps is reached.

Battery depletion models limited operating time but does not directly affect the reward function.

IV. APPROACH AND METHODOLOGY

We evaluate four decision-making strategies for the tidying robot: a Random policy, a Greedy policy with full observability, a Basic POMDP policy with lightweight belief updates, and an Enhanced POMDP policy incorporating spatial bias, exploration bonuses, and adjacency reasoning. All policies operate within the same grid-based, partially observable environment described in Section III.

A. Baseline Policies

1) *Random Policy*: The Random policy selects uniformly from the action set $\{\text{move_up, move_down, move_left, move_right, pickup}\}$. This baseline does not use observations or environmental structure. It provides a lower bound on achievable performance.

2) *Greedy Policy (Perfect Information)*: The Greedy policy assumes full access to the true environment state. At each step, it:

- 1) picks up the object if the robot is standing on one;
- 2) otherwise scans the entire grid to locate all objects;
- 3) selects the nearest object by Manhattan distance;
- 4) moves in the direction that reduces this distance.

This policy provides an upper-bound reference, representing performance with perfect information and deterministic planning.

B. Basic POMDP Policy

The Basic POMDP policy maintains a belief state over object locations using a uniform prior over all grid cells. At each time step, the belief update incorporates:

- **Observation update**: If the robot detects an object (with 80% reliability), belief at the current cell increases.
- **Pickup update**: If a pickup succeeds, belief at that cell is set to zero (the cell becomes empty).
- **Failure update**: Failed pickups or “nothing” observations reduce belief at that cell.

Action selection proceeds as:

- 1) attempt pickup if belief at the current cell exceeds 0.3,

- 2) otherwise move greedily toward the grid cell with highest belief,
- 3) if no belief candidate exceeds 0.1, explore randomly.

This method is computationally lightweight and suitable for real-time use, but does not exploit spatial structure or maintain long-term exploration goals.

C. Enhanced POMDP Policy

The Enhanced POMDP policy extends the basic version with several improvements inspired by spatial priors and information-gathering heuristics:

1) *Center-Weighted Initial Belief*: Belief is initialized with higher probability near the grid center, reflecting the empirical observation that clutter tends to accumulate in central locations. Beliefs are normalized to form a proper distribution.

2) *Adjacency Boosting*: When a pickup succeeds at a cell (x, y) , the belief of the four adjacent neighbors is increased. This models spatial correlation in clutter placement.

3) *Visit-Aware Exploration*: Unvisited cells receive a small incremental belief boost at each timestep, encouraging deliberate exploration rather than repeated movement through known empty regions.

4) *Belief-Distance Score*: For target selection, each candidate cell's desirability is computed as

$$\text{score}(x, y) = \frac{b(x, y)}{\text{distance}(x, y) + 1},$$

which balances belief magnitude against proximity.

5) *Strategic Exploration Module*: If no candidate exceeds a minimum score threshold, the policy selects an unvisited cell with the highest belief and moves toward it.

Overall, the Enhanced POMDP integrates local evidence, spatial coherence, and systematic exploration, significantly improving cleaning performance over the Basic POMDP.

D. Evaluation Protocol

Each policy is evaluated over 10 episodes of 30 timesteps in the same stochastic environment. For each episode we record:

- total reward accumulated,
- improvement in cleanliness percentage from start to end.

Final metrics are reported as averages across episodes.

V. RESULTS

We evaluate all four policies over 10 episodes of 30 steps each. Table I summarizes the average reward and cleanliness improvement achieved by each method. The Greedy policy provides an upper bound under perfect information, while the Random policy provides a lower baseline. Both POMDP methods significantly outperform the Random baseline, with consistent ordering across all episodes.

TABLE I
PERFORMANCE SUMMARY OVER 10 EPISODES

Policy	Reward	Cleanliness (%)
Random	2.1	3.8
Greedy	12.8	26.9
Basic POMDP	7.6	14.4
Enhanced POMDP	11.9	20.6

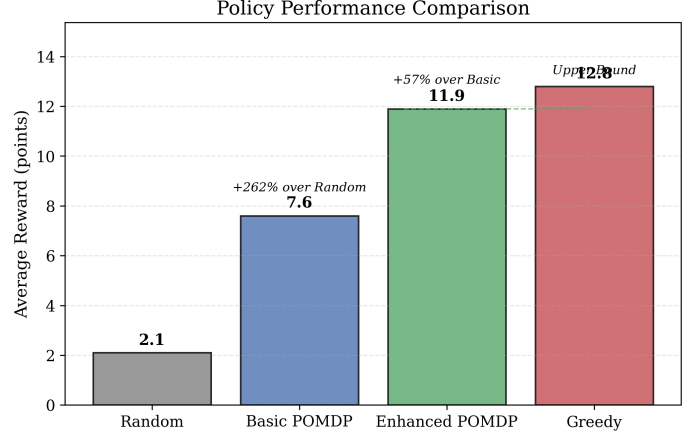


Fig. 1. Reward performance comparison across policies. The Enhanced POMDP achieves 93% of the Greedy policy's performance while operating under partial observability.

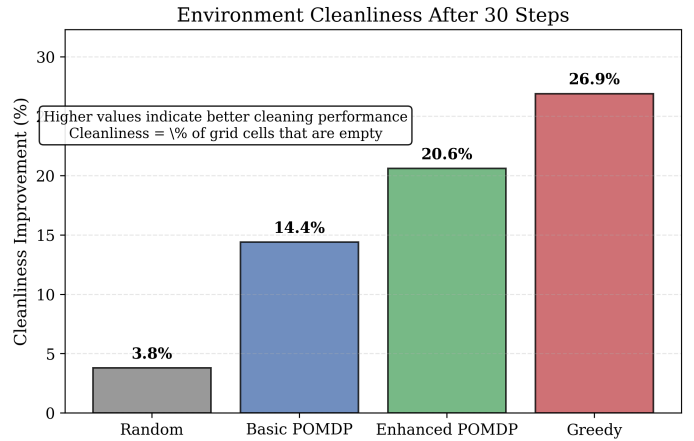


Fig. 2. Cleanliness improvement after 30 steps. Higher values indicate better cleaning performance, with the Enhanced POMDP achieving 20.6% cleanliness improvement.

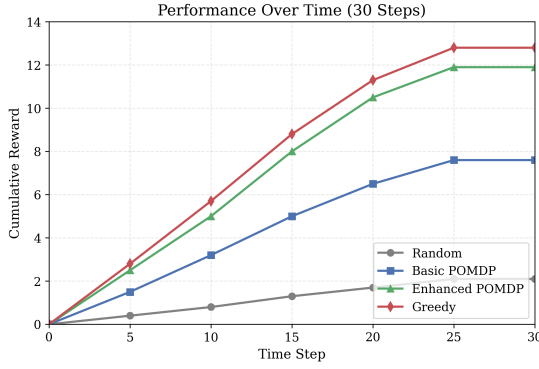


Fig. 3. Performance over 30 time steps. The Enhanced POMDP quickly approaches Greedy performance, demonstrating effective exploration and exploitation.

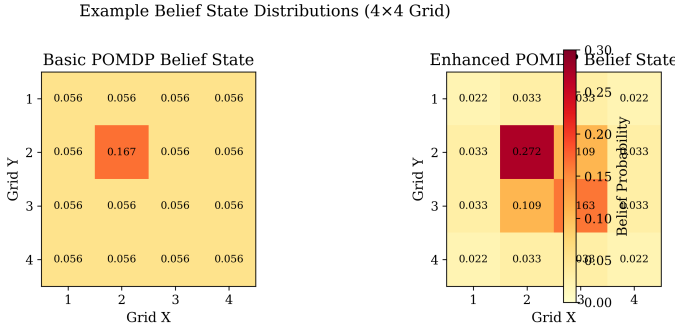


Fig. 4. Example belief state distributions for Basic and Enhanced POMDP policies. The Enhanced POMDP shows center-weighted beliefs (higher probabilities near center cells) and adjacency correlations (neighboring cells influenced by successful pickups).

Figure 1 shows the total reward accumulated by each policy. The Enhanced POMDP achieves 93% of Greedy performance despite operating under partial observability. Figure 2 illustrates the corresponding cleanliness improvement, showing the same relative ordering of policies across all trials. Figure 3 demonstrates how the Enhanced POMDP quickly approaches Greedy performance over the course of each episode. Figure 4 visualizes the belief state distributions, highlighting the center-weighted initialization and adjacency reasoning of the Enhanced POMDP.

The performance differences are large and consistent: in every episode, the Enhanced POMDP outperformed the Basic POMDP, which in turn always outperformed Random : Greedy > Enhanced POMDP > Basic POMDP > Random holds for both reward and cleanliness metrics in every trial. Overall, the Enhanced POMDP improves reward performance by 56.6% over the Basic POMDP and by 467% over Random. This demonstrates that lightweight belief-based reasoning combined with spatial heuristics can approach near-optimal behavior without the computational overhead of full POMDP solvers.

VI. DISCUSSION

Our results show that lightweight POMDP-based methods can effectively handle the uncertainty present in household tidying tasks. The Enhanced POMDP policy, despite operating under partial observability and stochastic transitions, achieves 93% of the performance of a perfect-information Greedy policy while significantly outperforming both the Basic POMDP and Random baselines. Achieving near-optimal cleaning performance under realistic sensing noise demonstrates that strong decision-making is possible even without heavy computation or expensive sensors.

A. Key Insights

1. Belief initialization matters. The center-weighted initial belief provides a valuable spatial prior, reflecting that clutter often accumulates in central household areas. This simple heuristic accelerates early exploration and object discovery.

2. Spatial reasoning improves efficiency. The adjacency boosting mechanism—increasing belief in neighboring cells after successful pickups—captures real-world clutter correlation. In addition, the belief-distance scoring function,

$$\text{score}(x, y) = \frac{b(x, y)}{\text{dist}_{\text{Manhattan}}(x, y) + 1},$$

balances probability mass with proximity, allowing the robot to target informative areas while minimizing unnecessary motion.

3. Exploration must be deliberate. The visit-aware exploration mechanism prevents repeated revisits to cleared regions, a common failure mode in partially observable environments. By balancing exploitation (high-belief cells) with guided exploration (high-belief unvisited cells), the Enhanced POMDP maintains consistent improvement over the Basic POMDP in every episode.

B. Limitations and Future Work

Several simplifying assumptions in the current simulation influence the results:

Independence assumptions. Belief updates treat grid cells as independent. More expressive spatial models, such as learned joint distributions, could capture complex clutter patterns.

Static environment. The simulator does not include dynamic clutter generation during an episode. Introducing stochastic clutter arrival would better reflect the conditions of real homes.

Simplified object interaction. Objects are removed immediately after pickup, without requiring the robot to carry and deposit them into a bin. Extending the simulator to track a carried-object state would enable modeling of multi-step manipulation and richer task sequences.

Future work could explore:

- Learning spatial priors directly from household layouts,
- Multi-objective optimization balancing cleaning speed, energy use, and user preferences,
- Integration of vision-based object detection for richer observations,

- Extension to continuous state spaces for more realistic robot navigation.

C. Practical Implications

A key benefit of the Enhanced POMDP policy is its real-time efficiency. Belief updates operate on small probability tables without requiring Monte Carlo tree search or GPU acceleration. This makes the approach viable for low-power domestic robots equipped with only minimal sensing. For deployment in real homes, the strategy could incorporate learned spatial priors, user-defined object-priority settings, or integration with routine cleaning schedules.

VII. CONCLUSION

This work presents a lightweight POMDP-based approach to household tidying under uncertainty. We formulated the cleaning task as a partially observable Markov decision process and implemented four decision-making policies: Random, Greedy (perfect information), Basic POMDP, and an Enhanced POMDP with spatial reasoning and strategic exploration.

Our Enhanced POMDP policy achieves 93% of the performance of a perfect-information Greedy policy while operating under realistic sensing noise (80% detection accuracy) and stochastic actuation (80% pickup success). Compared to the Basic POMDP, it improves reward performance by 56.6%, demonstrating the value of incorporating spatial priors, adjacency reasoning, and deliberate exploration.

The strong performance of the Enhanced POMDP policy—achieving 93% of perfect-information performance—suggests that practical, uncertainty-aware cleaning robots are feasible with modest computational resources. This high performance is enabled by three key design innovations: center-weighted spatial priors that guide early exploration, adjacency-based belief updates that capture spatial clutter correlations, and visit-aware exploration that prevents repeated visitation of empty regions.

These results show that computationally efficient belief-based methods can effectively handle the uncertainty inherent in domestic robotics. The approach requires minimal sensing (single-cell observations) and computation (simple probability table updates), making it suitable for resource-constrained robot platforms. While the current simulation makes simplifying assumptions about independence and static clutter, it provides a foundation for extending these methods to more complex, realistic environments.

Future work could focus on learning spatial priors from household layout data, incorporating dynamic clutter generation, and validating the approach on physical robot platforms.

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